

Heartfelt thanks to our amazing core staff and volunteers who made the Wrangell Mountains Center warm, welcoming, fun, delicious, musical, inspiring, educational, and functional this summer. Well done! *Pictured left to right and top to bottom: Joe Donohue, Allison Sayer, Tyler Boyes, Brita Mjos, David Conner, Andrea Sokolowski, Mark Vail, Lindsey Pepper, Ben Ciampa.* Thanks, too, to our numerous other volunteers (listed on pages 11-12) who were absolutely invaluable.



WMC Wishlist

If you would like to give new or used items that will directly enhance our effectiveness, we could really use the following items...

Satellite phone(s)	Bicycle trailers
Digital Projector	Canning jars
Pick Axe	Laptop Computer
Hatchet	Crampons and ice axes
Firewood	Nice pots and pans
Bicycles	Art supplies

To contribute any of the above items directly, or to make a donation toward the purchase of one of the above, contact jeremy@wrangells.org

Glaciology Wrangell Mountains Center-style

by Tim Bartholomaus

About 100 miles south-southeast of the old Hardware Store, at the hinge between Southcentral and Southeast Alaska, I study Yahtse Glacier. Yahtse ends in the ocean, and calves icebergs into remote Icy Bay, a crevice in Alaska's coastline tucked between two of the biggest glaciers in North America. At Yahtse, I and others from the University of Alaska at Fairbanks are trying to better understand iceberg calving, one of the ways in which the largest glaciers in the world lose mass.

Early this summer, as the fireweed in the Wrangells was just starting to bloom, we flew out of McCarthy, across the Chitina River plain and over the St. Elias Range. For four glorious days, we were blessed with rich, blue skies on the coast, and light winds, so the technical, instrumentation-based work we needed to do by helicopter went quickly. But rather than dashing back home after the active, adrenaline-infused work was complete, this time I was joined by two friends from the WMC for 10 days of focused observation of the glacier terminus. First David Conner, a Wrangell Mountains Center alum and intern, and then Jared Steyaert, former WMC facilities manager, joined me at our Yahtse camp, where together we watched the terminus for 81 hours, and recorded every crack, pop, crumble and topple that occurred as Yahtse's toe fell off into Icy Bay.

Our specific goal was to build a record of observed calving events that we could compare with data from seismometers. This database of terminus events will allow us to search for patterns of relative quiet and activity at the terminus, and to assess how tide cycles and rates of glacier flow might also influence the frequency of iceberg calving. After recording each of 4,800 events, we now have that database. But beyond that, I also developed, with David and Jared's help, a better intuition for calving—a more fundamental understanding of when and how ice falls inexorably from the glacier's terminal cliff. Through careful attention, we became able to predict when the most spectacular calving events would occur and why some parts of the terminus held together while other parts seemed to tumble continuously. Although technology allows for new measurements and knowledge that was not possible 50 or even five years ago, no fancy instrument could have replaced that which I gained from my time spent sitting, watching, pencil in hand, at the terminus of Yahtse Glacier. I think it was Jared who said, "This is *so* Wrangell Mountains Center." ∞